

MLFF Architecture Notes - Revision 57

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Gate: CUEQ-PHASE2

Revision summary

Revision 57 implements the optional CUEQ-PHASE2 selected-head source-execution/DATA6 qualification control plane while preserving the original six-head MH-1/`omat_pbe` checkpoint as scientific authority.

The gate locks the exact MH-1 SHA-256/source-potential digest and EXTRACT1 selected-head SHA-256/qualification digest. The candidate execution realization is content-addressed from those identities plus the CUEQ-DEP1 runtime, pure-CuEq mode, and dtype.

The deterministic development corpus is stratified across every declared-available composition/species, thermomechanical, high-force/difficulty, unusual local/mobile-ion, large/high-edge-count, and ordinary regime. Locked-test configurations cannot tune the realization or tolerances.

Energy/force/stress/descriptor acceptance reuses the existing MACE acceleration-parity authority. PHASE2 additionally requires difficulty parity, frozen-reference-transform PCA/FPS parity, exact DATA6/DATA7 selection fingerprints, and explicit scientific/execution lineage. Pseudolabel/E0 authorization requires separate value/E0 parity plus dual lineage.

Authorization boundary

A positive record may authorize the derived selected-head CuEq realization for source inference, DATA6, and source evaluation; pseudolabel generation requires the additional evidence above. It can never authorize direct CuEq execution of the original six-head MH-1 checkpoint or change generated defaults. CUEQ-PHASE1 remains the separate training-only authority; PERF-CERT1 remains the end-to-end certification authority.

Qualification state and next gate

Schemas, reducers, tamper guards, deterministic-corpus contract, deferred-state tooling, dependency graph, and FINAL-GPU1 handoff are CPU/control-plane qualified. Positive CUEQ-DEP1 plus PHASE2 accelerator evidence remains deferred to FINAL-GPU1. Preflight v4 carries PHASE1 and PHASE2 as independent deferred states.

The next implementation gate is **PERF-CERT1**. Its GPU execution remains deferred until the complete final release package is ready for the user's one-shot workstation qualification.